

Omar A. Khan

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Research Interests

I am broadly interested in **embodiment**, **interaction**, and **perception** in **XR (AR, VR, MR)** environments, with a focus on avatars, virtual agents, social interaction, haptic perception, body perception, and tangible/physical user interfaces, all within the larger scope of **HCI**.

Education

PhD	Drexel University , Computer Science Advisor: Dr. Tiffany D. Do	Sep 2025 - present Philadelphia, PA, USA
BSc	University of Calgary , Computer Science (First-Class Honours) Advisor: Dr. Kangsoo Kim	Sep 2021 - Jun 2025 Calgary, AB, Canada

Publications

Peer-Reviewed Journal Articles

- J1.** Omar A. Khan, Bao Han Trinh, Lee Lisle, Tiffany D. Do. "Having Dog Ears 'for Real': Effects of Active and Passive Haptics on Embodying Non-Human Body Parts in VR". To appear in IEEE Transactions on Visualization and Computer Graphics (TVCG), 2026. <https://doi.org/10.48550/arXiv.2606.26364>
Acceptance rate: 16% • Impact Factor: 6.8
- J2.** Omar A. Khan, Junyeong Kum, Hyeongil Nam, Myungho Lee, Kangsoo Kim. "Influence of Avatar Appearance and Target Distance on Locomotion Method Selection in Virtual Reality". IEEE Transactions on Visualization and Computer Graphics (TVCG), vol. 32, no. 5, pp. 4668-4677, 2026. <https://doi.org/10.1109/TVCG.2026.3679092>
Acceptance rate: 20% • Impact Factor: 6.8
- J3.** Omar Khan, Hyeongil Nam, Kangsoo Kim. "Impact of Avatar-Locomotion Congruence on User Experience and Identification in Virtual Reality". IEEE Transactions on Visualization and Computer Graphics (TVCG), vol. 31, no. 11, pp. 9878-9888, 2025. <http://doi.org/10.1109/TVCG.2025.3616836>
Acceptance rate: 8% • Impact Factor: 6.8 • 🏆 Honorable Mention Award at IEEE ISMAR 2025 (Top 3%)

Peer-Reviewed Extended Abstracts and Posters

- E1.** Zaid Ahmed, Omar A. Khan, Hyeongil Nam, Kangsoo Kim. "Exploring Experiential Differences Between Virtual and Physical Memory-Linked Objects in Extended Reality". *Proceedings of ACM CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA)*, no. 269, pp. 1-5, 2026. <https://doi.org/10.1145/3772363.3798977>
Acceptance rate: 38%
- E2.** Omar Khan, Zaid Ahmed, Hyeongil Nam, Kangsoo Kim. "TangibleMoments: Embedding XR Memories onto Physical Objects". *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1142-1146, 2025. <https://www.doi.org/10.1109/VRW66409.2025.00227>
- E3.** Omar Khan, Anh Nguyen, Hyeongil Nam, Kangsoo Kim. "Investigating Visual Guide Cues in VR: Impacts of Virtual Humans and Symbol-Based Navigation on Real-World Performance and Experience". *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 504-509, 2025. <https://www.doi.org/10.1109/VRW66409.2025.00110>
- E4.** Omar Khan, Hyeongil Nam, Kangsoo Kim. "I look like a gorilla, but don't move like one!": Impact of Avatar-Locomotion Congruence in Virtual Reality". *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1202-1203, 2025. <https://www.doi.org/10.1109/VRW66409.2025.00247>

- E5. Omar Khan**, Anh Nguyen, Michael Francis, Kangsoo Kim. “Exploring the Impact of Virtual Human and Symbol-Based Guide Cues in Immersive VR on Real-World Navigation Experience”. *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 883-884, 2025. <https://www.doi.org/10.1109/VRW62533.2024.00238>

Awards

- 2025 **Best Paper Award Honorable Mention (Top 3%)**, IEEE International Symposium on Mixed and Augmented Reality (ISMAR)
- 2025 **IDEAS Travel Fund**, University of Calgary - \$1352
- 2024 **Undergraduate Research Symposium Faculty of Science Award**, UCalgary Students' Union - \$1000
- 2024 **Summer Research Studentship Award**, Alberta Innovates - \$8500
- 2024 **Graeme Bell Travel Award**, University of Calgary - \$1350
- 2023 **Undergraduate Student Research Award**, Natural Sciences and Engineering Research Council of Canada - \$7500
- 2022 **Jason Lang and Alexander Rutherford Scholarships**, Government of Alberta - \$5500

Academic Service & Outreach

Peer Review

- 2026 **IEEE ISMAR, ACM VRST**
- 2025 **ACM VRST**

Student Volunteer

- 2025 **IEEE ISMAR**

Outreach

- 2026 **Drexel CCI Doctoral Student Association**, Event Coordinator

Invited Talks

- 2026 **Stony Brook University (CSE 566)**, Modern Research in Virtual Embodiment